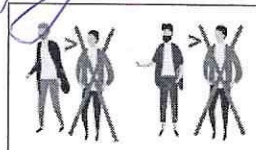
Как найти жену если я сам Султанов



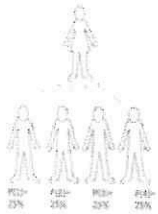
Say **NO** to the **first**

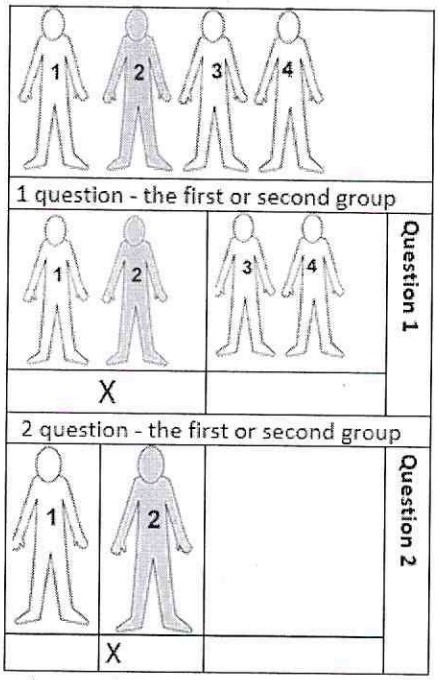


Say **YES** to the **second** if it is better than the first



Say **NO** to the **third** only if it is worse than all the others





Average number of questions =

2*0.25+ 2*0.25+ 2*0,25+ 2*0,25 =2

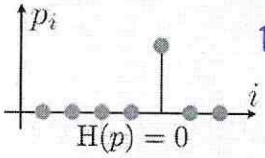
Average number of questions =

1*0.5+	2*0.25+	3*0,125+	3*0,125
			

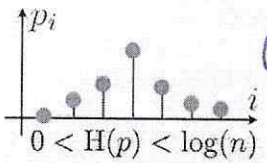
Question 1. Is this Zuckerberg?	 50%	1*0.5
Question 2. Is this Sergey Brin?	 25%	2*0.25
Question 3. Is this Stefan from BMW?	 12,5%	3*0,125
So Prince Saud	 12,5%	3*0,125
Average number of questions =1,75		

$$H(X) = \sum_{i=1}^{n} p(x_i) \log_b \frac{1}{p(x_i)}$$

$$\sum_{i=1}^{n} p(i) \log_2 \frac{1}{p(i)}$$

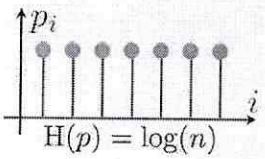


Quantifying information

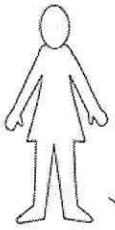


$$I(x_i) = \log_2\left(\frac{1}{p_i}\right)$$

number of bits required to encode choice

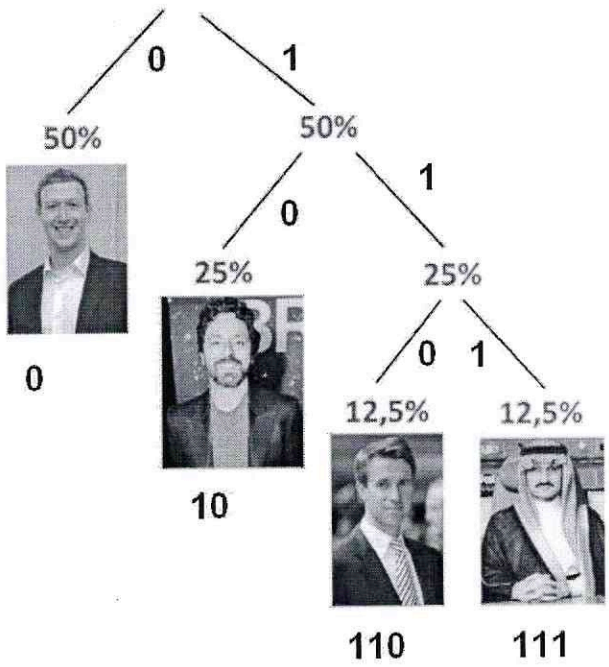


$$\sum_{i=1}^{n} p(x_i)\, I(x_i)$$



50 billion	25 billion	12.5 billion	12.5 billion
			
Mark Zukerberg P(1)= 50%	Sergey Brin P(2)= 25%	Stefan Quandt P(3)= 12,5%	Prince Al Saud P(4)= 12,5%

0.1



First-order approximation (symbols independent but with frequencies of Belarusian txt).

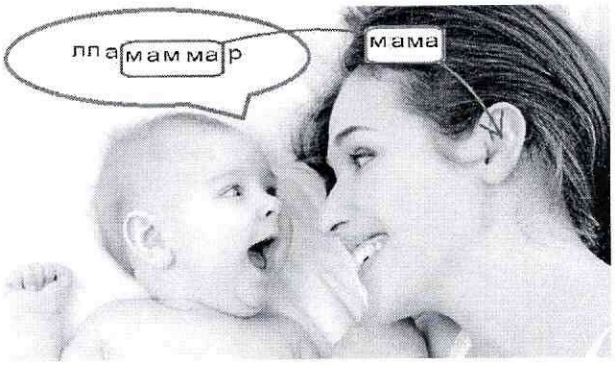
Мама мыла ра

М - 3	30%	1-3	М
а - 4	40%	4-7	а
ы -1	10%	8	-ы
л - 1	10%	9	-л
р - 1	10%	10	-р

10

лламаммар

Автор звуков



Мама мыла ра

Ма - 2	22%	1-2	ма
ам - 2	22%	3-4	ам
мы - 1	11%	5	мы
ыл - 1	11%	6	ыл
ла - 1	11%	7	ла
ар - 1	11%	8	ар
ра - 1	11%	9	ра

9

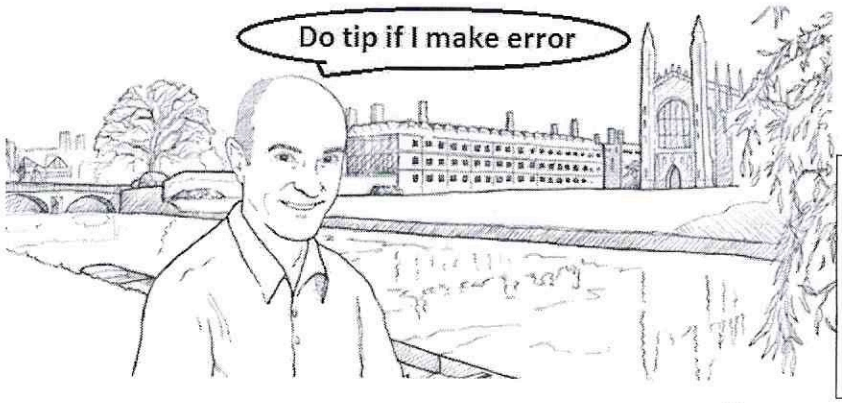
0. 4 6 7 3 1 9 1 6 7 3 5

ам ыл ла ам ма ра ма ыл ла ам мы

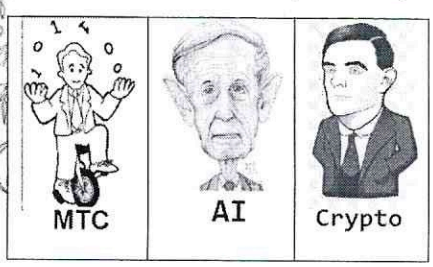
мылла рама



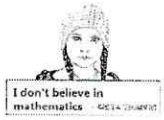
Second-order approximation (digram (2-symbols) structure as in Belarusian)

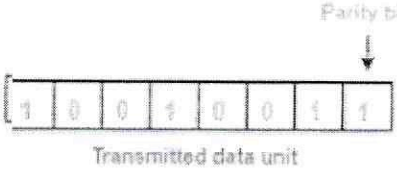


Sir **Dr. D. MacKay**,
University of Cambridge
(22 April 1967 – 14 April 2016)

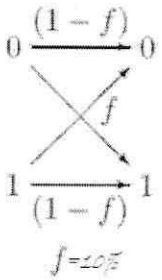


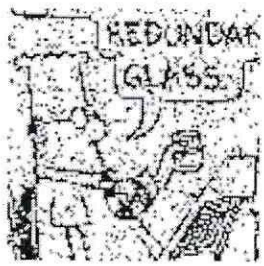
"I believe in clean energy,
but I also believe in mathematics"

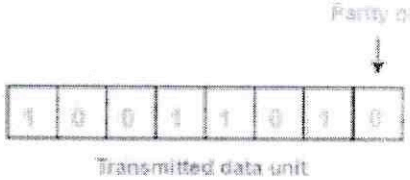


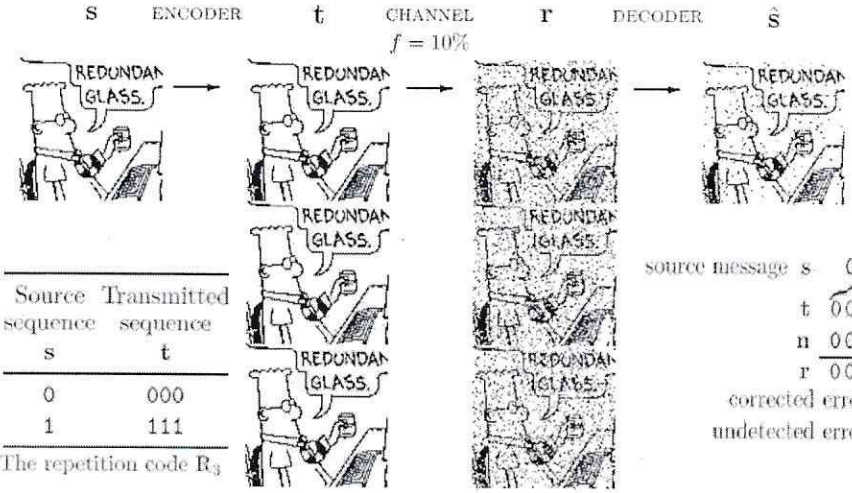










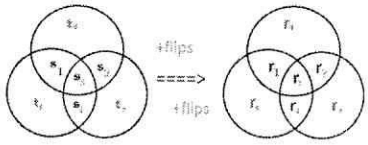


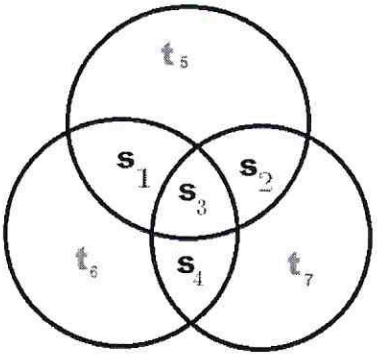
Source sequence s	Transmitted sequence t
0	000
1	111

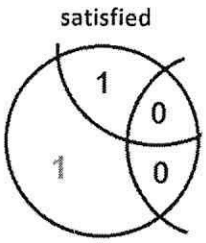
The repetition code R_3

source message s	0	0	1	0	1	1	0
t	000	000	111	000	111	111	000
n	000	001	000	000	101	000	000
r	000	001	111	000	010	111	000
corrected errors		*					
undetected errors					*		

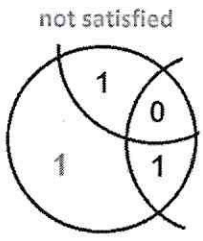
7.4. Hamming code. $\frac{4}{\Sigma} \rightarrow \frac{7}{t}$



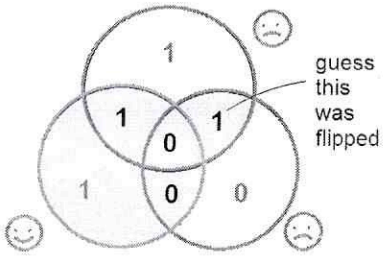




1+1+0+0=**0**



1+1+0+1=**1**

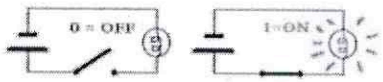


Massachusetts Institute of Technology (MIT)



Lecture by Pr. Bob Gallagher
Boole (1815-1864) & **Shannon** (1916-2001)



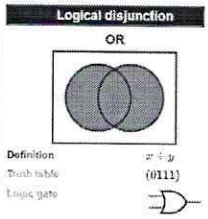




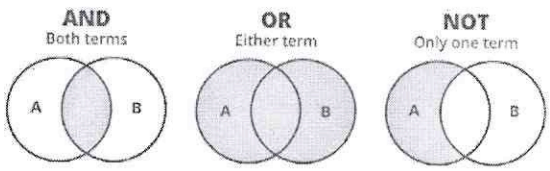
Logical addition (disjunction)

A	B	F=A∨B
0	0	0
0	1	1
1	0	1
1	1	1

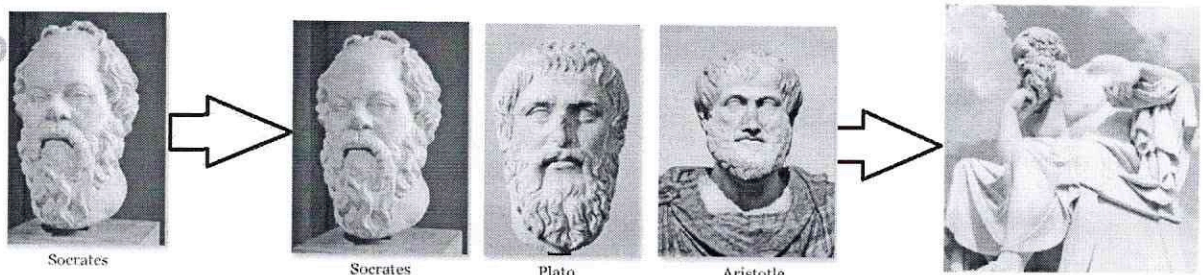
A	B	$A \vee B$
True	True	True
True	False	True
False	True	True
False	False	True



BOOLEAN LOGIC



Good logic



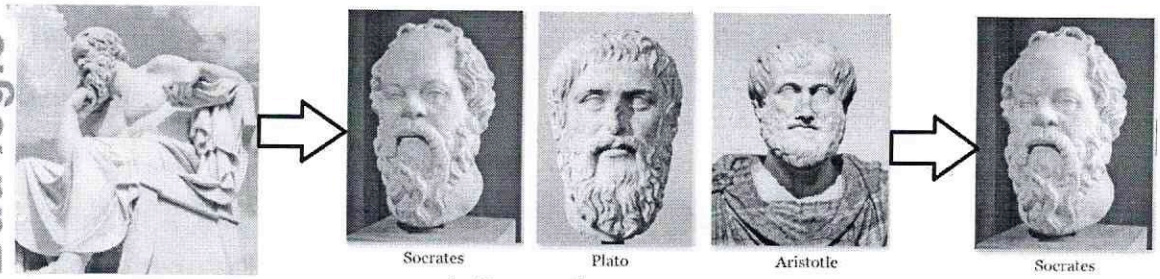
Socrates was a philosopher

philosophers are men

Socrates was a man



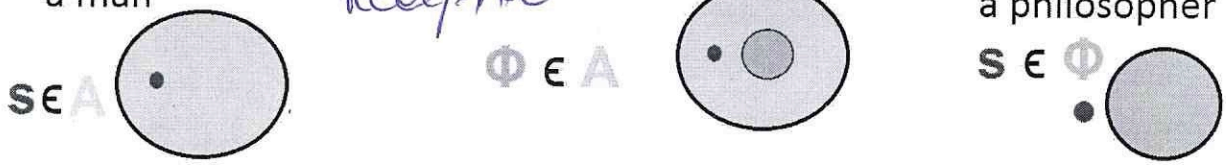
Bad logic

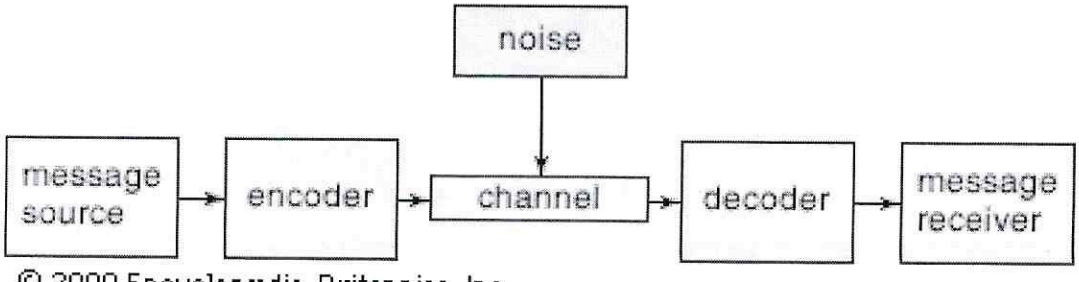


Socrates was a man

philosophers are men

Socrates was a philosopher





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Resume of Lecture by Pr. Bob Gallagher from MIT

Massachusetts Institute of Technology (MIT)

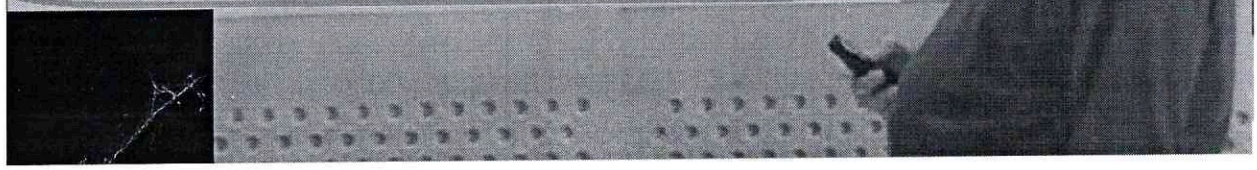
George Boole (1815-1864) developed Boolean logic

The principles of logical thinking have been understood (and occasionally used) since the Hellenic era.

Boole's contribution was to show how to systemize these principles and express them in equations (called Boolean logic or Boolean algebra).

Claude Shannon (1916-2001) showed how to use Boolean algebra as the basis for switching technology. This contribution systemized logical thinking for computer and communication systems, both for the design and programming of the systems and their applications.

Logic continues to be abused in politics, religion, and most non-scientific areas.





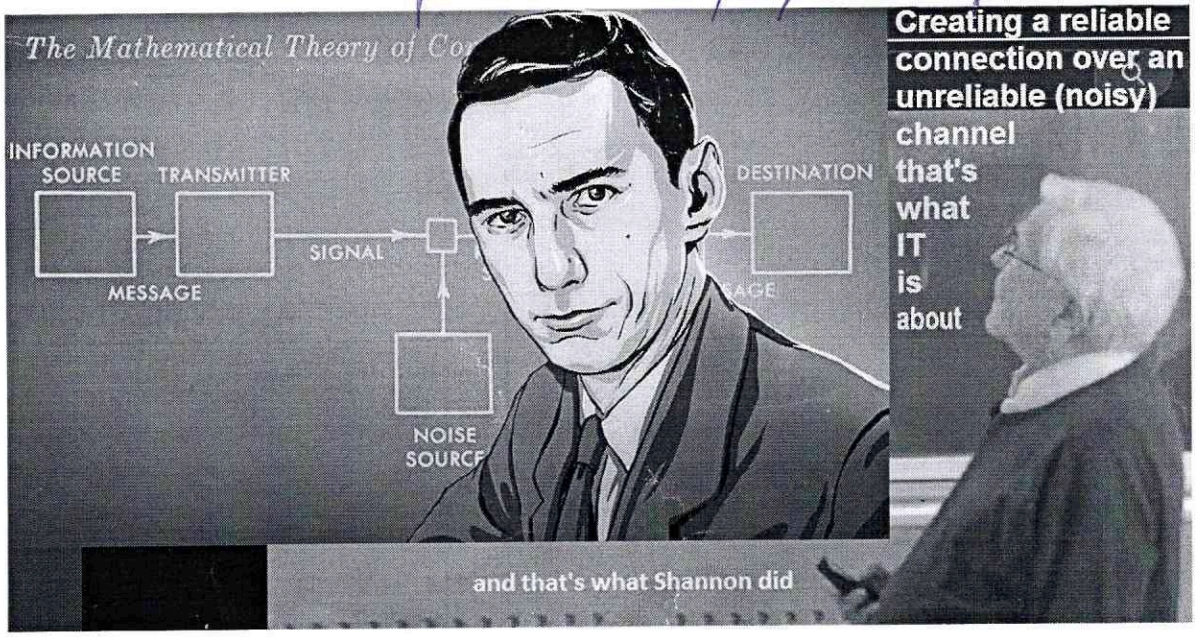
A little nationalistic, but this is an sample of right logic

Kant, Gauss, Goethe are great → Kant, Gauss, Goethe - Germans → Germany Great

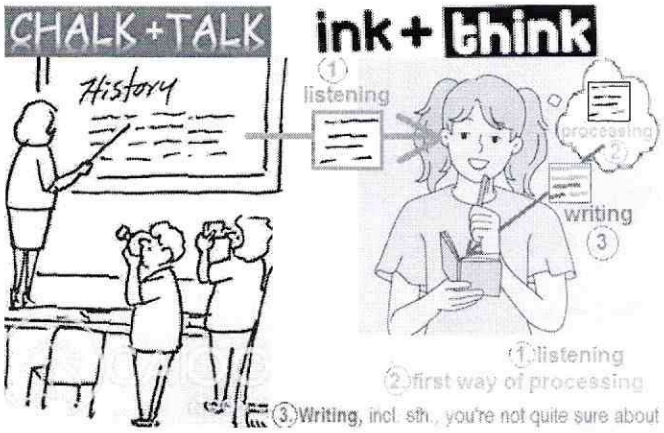


Bad logic (abuse of logic)

Всё ещё теория информации

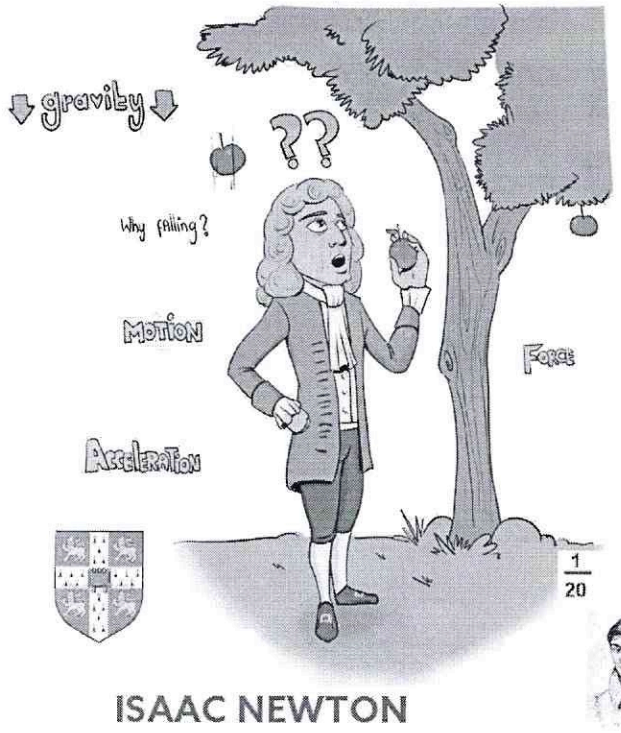


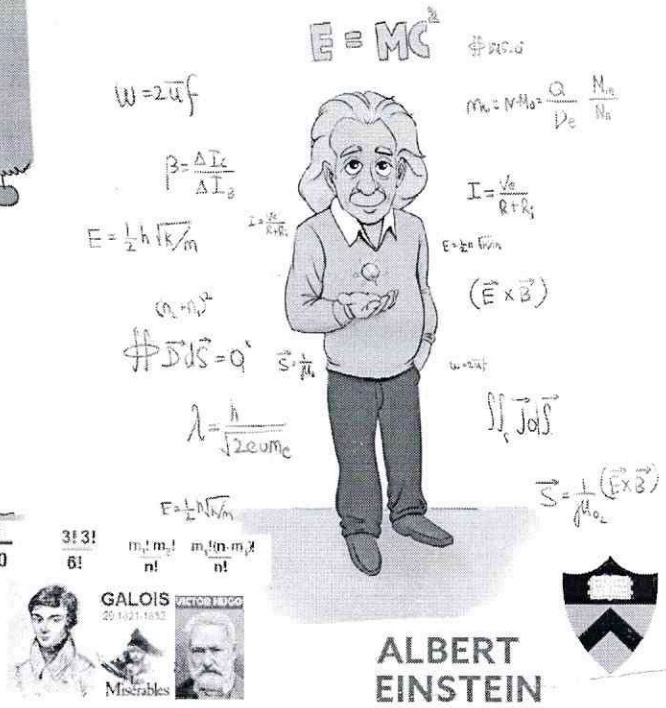


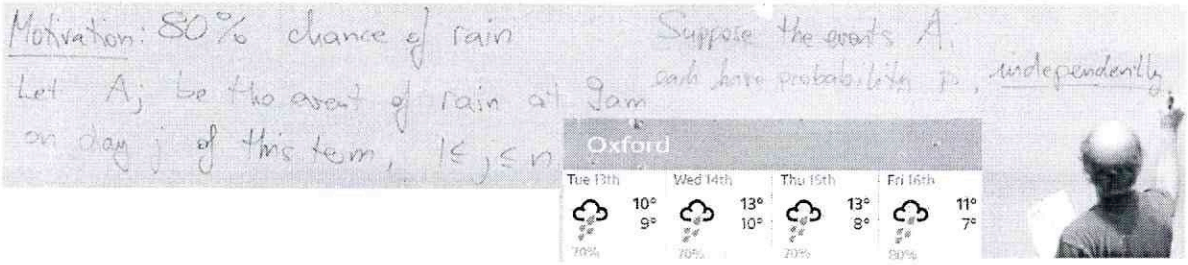


School ↓gravity↓ MOTION ==formalism==> University $E=MC^2$ $\oint \vec{J} d\vec{S}$

CONCRETE AND ABSTRACT THINKING







Markoff Chain Probability Model for Oxford Weather

